Real-Estate Price Prediction and Price-Tier Classification

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MLE-AI-201 Machine Learning – AI Academy, Cohort II

18.10.2026

Problem and project goals

Task A: regression

- Predict the advertised property price in AZN.
- Exact price is difficult because listings are heterogeneous.

Task B: classification

- Derive a premium/standard label.
- Premium means price above the training median: **195,000 AZN**.

What we built

- Decision tree from scratch
- Pegasos SVM from scratch
- Random Fourier Feature SVM
- Ridge, SVR, and random-forest bonuses
- Matched scikit-learn baselines

Dataset, preprocessing, and leakage

- 100,775 listings; duplicate rows were removed before modelling.
- 167 encoded features.
- Prices and areas parsed from Azerbaijani text and units.
- Price outliers filtered with the 1.5 IQR rule.
- Missing numeric values imputed; missing categories retained.

Main features

Latitude/longitude, area, rooms, land area, floor, total floors, renovation, city, location, and property category.

Encoding

Binary fields become 0/1 indicators; multi-category fields use one-hot columns. Missing indicators are preserved where meaningful.

Split

41,787 train / 8,955 validation / 8,955 test

Leakage columns removed

`unit_price` and `total_price` are target-derived and would reveal the answer. IDs, URLs, descriptions, scrape metadata, and duplicate aliases were also excluded.

Cleaning and split: The shared cleaning pipeline runs before splitting. Encoding, imputation, scaling, and the tier threshold are training-derived.

Decision tree from scratch

At each node, the tree searches feature thresholds and chooses the largest impurity decrease:

$$\Delta I = I_{\text{parent}} - \frac{n_L I_L + n_R I_R}{n_L + n_R}.$$

- Classification: Gini or entropy
- Regression: MSE reduction
- Leaf: majority class or mean price

Selected settings

- Depth: 6
- Minimum leaf: 15
- Classification criterion: entropy

Why it matters

Depth 6 captures nonlinear interactions without an unrestricted tree. Leaf means are stable, but can underpredict unusual luxury properties.

Pegasos SVM from scratch

We minimise the regularised hinge-loss objective:

$$J(\mathbf{w}, b) = \frac{\lambda}{2}\|\mathbf{w}\|^2 + \frac{1}{n}\sum_i \max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)).$$

- Labels are converted from {0,1} to {-1,+1}.
- Features are standardised using training statistics.
- One training example is sampled per iteration.
- Learning rate: $\eta_t = 1/(\lambda t)$.
- The nonlinear extension maps features with 100 Random Fourier Features, approximating an RBF kernel.

Additional baselines

Ridge uses regularised least squares; SVR ignores residuals inside an epsilon tube; the random forest averages depth-limited trees trained on bootstrap samples.

Experimental setup

Training protocol

- Fixed seed: 42
- Validation used for hyperparameter selection
- Test set used once for final scoring
- Three-fold check on the original training partition

Regression metrics

- RMSE, MAE, R^2

Classification metrics

- Precision, recall, F1, ROC-AUC

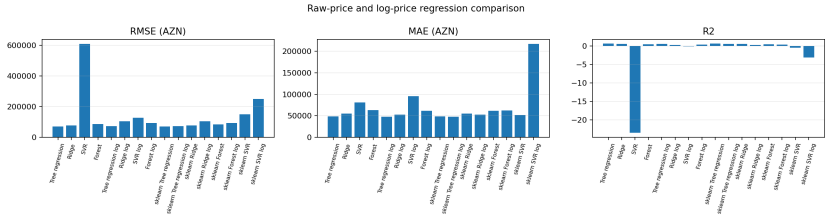
Comparisons

- Custom implementations
- Matched scikit-learn tree, Ridge, forest, SVM, and SVR
- Separate raw-price and $\log(1 + \text{price})$ regressors

Log models were trained separately on $\log(1 + \text{price})$ and converted back to AZN with expm1 before scoring.

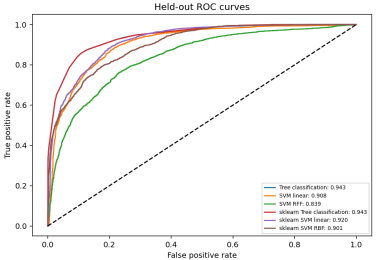
Regression results

Model	RMSE (AZN)	MAE (AZN)	R^2
Custom raw tree	**71,741**	48,559	**0.660**
Custom log tree	74,011	**48,313**	0.639
Custom Ridge	78,566	54,985	0.593
Custom forest	86,455	62,943	0.507
sklearn tree	71,735	48,535	0.660
sklearn forest	84,442	61,873	0.530



Classification results

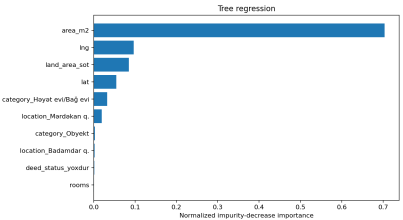
Model	Precision	Recall	F1	ROC-AUC
Custom tree	**0.874**	0.873	**0.873**	**0.943**
Linear Pegasos	0.798	**0.888**	0.841	0.908
RFF Pegasos	0.751	0.784	0.767	0.839
sklearn tree	0.872	0.873	0.873	0.943
sklearn linear SVM	0.817	0.874	0.845	0.920



What worked, what failed, and why

Strengths

- Tree classifier separates premium and standard listings well.
- Custom tree nearly matches the sklearn tree.
- Raw and log trees give consistent conclusions: raw RMSE 71,741 AZN, $R^2 = 0.660$; log RMSE 74,011 AZN, $R^2 = 0.639$.



Limitations

- Exact price errors remain large: MAE 48,559 AZN.
- Errors rise for expensive homes; properties above 500,000 AZN average about 165,000 AZN error.
- Advertised prices are noisy, quality details are missing, and SVR was unstable: RMSE 609,574 AZN, $R^2 = -23.518$.

Contributions and next steps

Gunel Garayzade	Evaluation orchestration, metrics, experiments, and result analysis
Jale Afandiyeva	SVM and linear-baseline work, including Pegasos and SVR comparisons
Kheyransa Grochuyeva	Decision-tree and random-forest implementation and testing
Nazrin Burziyeva	Data preparation, integration, reproducibility, report, and presentation

Next steps

- Add safer interaction features and richer location features, such as distance to metro and neighbourhood context.
- Test nonlinear tabular models such as gradient boosting, with more extensive tuning.
- Model high-value properties separately.
- Report relative errors and prediction intervals.

Main finding

The available listing features contain meaningful signal. The custom tree achieved strong price-tier classification and meaningful, but not precise, price prediction.

- Best raw regression: RMSE 71,741 AZN, $R^2 = 0.660$.
- Best custom classifier: F1 0.873, ROC-AUC 0.943.
- The system is suitable for an academic approximate-estimation task, not professional valuation without richer data and uncertainty estimates.

Thank you